

# CORE: Mitigating Majority Bias in Language Agents via Counterfactual Utility Reweighting

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## Abstract

Current large language models (LLMs) and autonomous agents are optimized primarily to match historical data distributions, either through behavioral cloning or preference alignment objectives. Consequently, during multi-step reasoning and planning, these models exhibit a pronounced *majority-frequency bias*: they consistently favor paths that are highly probable under the training distribution, even when those paths lead to sub-optimal outcomes or cognitive failures. Crucially, in complex, long-horizon tasks, breakthrough solutions and highly effective strategies often reside in the tail of the distribution—representing low-frequency but high-utility trajectories. In this paper, we formalize the problem of distinguishing high-frequency, low-outcome trajectories from low-frequency, high-outcome trajectories. We propose **CORE** (Counterfactual Outcome-Regularized Estimation), a principled framework that moves language agents from distribution matching to outcome optimization. CORE introduces a latent semantic decision abstraction to handle token-level sparsity, estimates the causal utility of trajectories via inverse-propensity-guided counterfactual evaluation, and applies an epistemic value-gating mechanism to ensure that the agent safely reverts to majority tracks when they are optimal. Across extensive evaluations on mathematical reasoning (MATH, GPQA, AIME) and long-horizon agent execution (WebArena, SWE-bench, GAIA), CORE demonstrates substantial improvements, successfully uncovering hidden high-utility strategies without degrading performance on standard tasks.

## 1 Introduction

Autoregressive large language models (LLMs) are structurally bound to maximize the conditional likelihood of tokens given historical prefixes:  $P_\theta(x_t|x_{<t})$ . When fine-tuned using Reinforcement Learning from Human Feedback (RLHF) or Direct Preference Optimization (DPO), their target remains tightly coupled to a reference policy or human preference distribution, ensuring that the model mimics the common consensus or high-frequency behaviors present in the training corpora [33, 35]. While this distribution-matching paradigm delivers fluent generation and strong common-sense capabilities, it induces a foundational failure mode in complex reasoning and long-horizon agent planning: **majority-frequency bias**.

Majority-frequency bias occurs when an agent encounters a critical decision node and selects a path that is overwhelmingly common in its training distribution, despite that path leading to catastrophic failure or a dead end in the specific test instance. Conversely, innovative problem-solving, optimal edge-case handling, and revolutionary logical inferences typically appear as anomalies or low-frequency paths within broad behavioral distributions [29, 39]. For instance, in advanced mathematics or highly specialized software engineering, the standard, intuitive approach (the majority trajectory) may fail due to subtle boundary conditions that require a rare, counterintuitive counter-strategy (the minority trajectory).

Existing test-time scaling methods, such as Self-Consistency [44] or standard Tree-of-Thought search [46], fail to alleviate this bias. Self-Consistency explicitly leverages majority voting, which inherently suppresses correct but rare solutions. Process Reward Models (PRMs) [14, 29] attempt to evaluate individual steps, but because their training distributions are heavily confounded by the base policy’s proposal frequencies, they systematically overrate high-frequency, plausible-sounding steps while penalizing unorthodox but correct ones.

To solve this, we must transition language agents from *distribution matching* toward *outcome optimization*. This requires addressing three fundamental challenges: 1) **Token-Level Sparsity**: In open-ended text, every unique string sequence has a low nominal frequency, making it hard to identify structural "minority" strategies based on raw tokens. 2) **Selection Bias & Confounding**: The historical training data confounds trajectory frequency with trajectory utility, requiring true counterfactual reasoning to isolate the causal impact of a rare reasoning step. 3) **Contrarian Degradation**: Over-correcting for majority bias could lead an agent to chase bizarre, low-probability errors when the common, high-frequency track is already perfectly correct.

**Our Contributions.** In this paper, we present **CORE** (Counterfactual Outcome-Regularized Estimation), a framework designed to robustly surface and exploit low-frequency, high-outcome reasoning paths:

- We formalize the majority-frequency bias problem in language space using a structural causal framework, rigorously separating trajectory generation probability from true future outcome utility.
- We introduce **Latent Semantic Decision Nodes (LSDNs)**, mapping continuous token sequences to a discrete, functional abstraction layer to measure structural path frequencies accurately.
- We design an **Epistemic Value-Gating (EVG)** objective that dynamically scales our counterfactual utility reweighting operator based on the statistical significance of the utility advantage, preventing performance degradation when the majority path is optimal.
- We conduct large-scale experiments across six challenging benchmarks (MATH, GPQA, AIME, WebArena, SWE-bench, GAIA). CORE consistently outperforms state-of-the-art test-time scaling methods (including DeepSeek-R1-style search and heavy PRMs), showing its greatest gains precisely when historical distributions are highly deceptive.

## 2 Background and Related Work

**Language Models as Policy Trajectories.** Formally, let a language model be a policy  $\pi_\theta(a|s)$ , where the state  $s_t \in \mathcal{S}$  represents the textual prompt and all tokens generated up to step  $t$ , and the action  $a_t \in \mathcal{A}$  represents the next token or a bounded sub-sequence (e.g., a line of reasoning or a tool call). A complete reasoning trajectory is denoted as  $\tau = (s_0, a_0, s_1, a_1, \dots, s_T) \in \mathcal{T}$ . Traditional pre-training and supervised fine-tuning (SFT) minimize the cross-entropy loss against a data distribution  $\mathcal{D}_{\text{data}}$ , optimizing the forward KL-divergence  $D_{\text{KL}}(\mathcal{D}_{\text{data}}||\pi_\theta)$ , which forces  $\pi_\theta$  to allocate probability mass proportionally to historical frequency.

**Test-Time Refinement and Search.** To bypass the limitations of greedy decoding, recent paradigms scale test-time compute. Self-Consistency [44] samples multiple trajectories  $\tau_i \sim \pi_\theta$  and uses a majority vote over final answers:  $\arg \max_y \sum_i \mathbb{I}(y_i = y)$ . While effective for reducing

stochastic variance, it exacerbates systemic majority biases. Tree-of-Thought (ToT) [46] and Monte Carlo Tree Search (MCTS) agents [20, 41] utilize value models to guide exploration. However, these value models are typically trained on data rolled out by  $\pi_\theta$  itself, making them highly susceptible to selection bias where unchosen or low-probability paths have poorly calibrated value estimates.

**Offline Reinforcement Learning in Alignment.** Value estimation under severe policy shift is a core challenge in offline RL [28]. Algorithms like Conservative Q-Learning (CQL) [27] or Implicit Q-Learning (IQL) [26] deliberately penalize out-of-distribution actions to maintain policy stability. When applied to language models via Direct Preference Optimization (DPO) [35] or Kahneman-Tversky Optimization (KTO) [17], they enforce strict regularizers ( $\mathbb{E}_\tau[D_{\text{KL}}(\pi_\theta||\pi_{\text{ref}})]$ ) that deliberately punish low-frequency trajectories under the reference distribution. This directly conflicts with the need to extract rare, high-performance reasoning paths that break away from common human errors.

### 3 Problem Formulation

Let  $\mathcal{T}$  be the space of all possible reasoning trajectories for a given context or task. Let  $U : \mathcal{T} \rightarrow \mathbb{R}$  be a ground-truth utility function (e.g., binary correctness of a mathematical proof or execution success of an agent action). The historical generation policy  $\pi_{\text{ref}}(\tau)$  represents the reference distribution inherited from large-scale pre-training and standard alignment.

We define two critical, latent sub-spaces of trajectories for a given context:

**Definition 1** (High-Frequency, Low-Outcome Space). *The high-frequency, low-outcome trajectory subspace  $\mathcal{T}_{\text{HF-LO}}$  satisfies:*

$$\mathcal{T}_{\text{HF-LO}} = \{\tau \in \mathcal{T} \mid \pi_{\text{ref}}(\tau) > \delta_H \quad \text{and} \quad \mathbb{E}[U(\tau)] < \epsilon_L\} \quad (1)$$

where  $\delta_H$  is a high-frequency threshold and  $\epsilon_L$  is a low-utility bound.

**Definition 2** (Low-Frequency, High-Outcome Space). *The low-frequency, high-outcome trajectory subspace  $\mathcal{T}_{\text{LF-HO}}$  satisfies:*

$$\mathcal{T}_{\text{LF-HO}} = \{\tau \in \mathcal{T} \mid \pi_{\text{ref}}(\tau) < \delta_L \quad \text{and} \quad \mathbb{E}[U(\tau)] > \epsilon_H\} \quad (2)$$

where  $\delta_L$  is a low-frequency threshold ( $\delta_L \ll \delta_H$ ) and  $\epsilon_H$  is a high-utility bound ( $\epsilon_H \gg \epsilon_L$ ).

Standard decoding objectives select trajectories based on their posterior probability under  $\pi_{\text{ref}}$ . Consequently, the probability of selecting an erroneous path from  $\mathcal{T}_{\text{HF-LO}}$  over an optimal path from  $\mathcal{T}_{\text{LF-HO}}$  is lower-bounded by the ratio of their reference priors:

$$\frac{P_{\text{select}}(\tau \in \mathcal{T}_{\text{HF-LO}})}{P_{\text{select}}(\tau \in \mathcal{T}_{\text{LF-HO}})} \propto \frac{\pi_{\text{ref}}(\mathcal{T}_{\text{HF-LO}})}{\pi_{\text{ref}}(\mathcal{T}_{\text{LF-HO}})} > \frac{\delta_H}{\delta_L} \gg 1 \quad (3)$$

Our primary objective is to design an optimized decision policy  $\pi^*(\tau)$  that prioritizes trajectories based strictly on their future utility  $U(\tau)$ , decoupling the final choice from the historical prior  $\pi_{\text{ref}}(\tau)$ , while satisfying a safety constraint that guarantees non-degradation when the majority path is already optimal (i.e., when  $\tau^* \in \mathcal{T}_{\text{HF-HO}}$ ).

## 4 Methodology

The **CORE** framework consists of three architectural components: (1) Abstracting continuous token steps into Latent Semantic Decision Nodes to overcome string-level sparsity; (2) An inverse-propensity score counterfactual utility estimator; and (3) An Epistemic Value-Gated decoding and optimization objective.

### 4.1 Theoretical Motivation

Consider a structural causal model (SCM) describing language generation, where  $C$  is the task context,  $Z$  is the underlying core reasoning strategy (latent),  $\tau$  is the rendered token-level trajectory, and  $Y = U(\tau)$  is the final outcome success. The historical training data creates a strong confounding link between  $C$  and  $Z$  based on human cognitive biases or common internet patterns. We wish to estimate the counterfactual outcome if we break this link and force the model to evaluate an alternate strategy:  $P(Y|\text{do}(Z = z))$ .

By applying the back-door adjustment formula over the context distribution, we isolate the true causal utility of a reasoning path. However, since we work in an offline setting or with a fixed reference policy  $\pi_{\text{ref}}$ , a naive value model will overfit to the observed correlations. To correct for this selection bias, we must reweight our trajectory valuations using the inverse of the propensity score—which in this case is exactly the conditional generation probability under the reference policy  $\pi_{\text{ref}}$ .

### 4.2 Latent Semantic Decision Nodes (LSDNs)

Because natural language is highly expressive, two trajectories can be identical in their mathematical or logical substance while having completely disjoint token sequences. Computing the frequency of a raw text trajectory  $\tau$  results in severe data sparsity ( $\pi_{\text{ref}}(\tau) \rightarrow 0$  for all long paths).

To solve this, we define a mapping function  $\Psi : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{Z}$ , where  $\mathcal{Z}$  is a discrete, low-dimensional space of semantic decision abstractions. We implement  $\Psi$  via a lightweight auxiliary freeze-frame encoder trained to predict future state trajectories. The structural frequency of a reasoning path step is thus computed over its latent trajectory  $\mathbf{z} = (z_0, z_1, \dots, z_K)$ , where  $z_k = \Psi(s_k, a_k)$ . The reference probability of a latent strategy is given by:

$$\Pi_{\text{ref}}(\mathbf{z}) = \prod_{k=0}^K P_{\text{ref}}(z_k \mid z_{<k}, C) \quad (4)$$

### 4.3 Counterfactual Utility Estimation and Objectives

We train a Counterfactual Value Network  $V_\phi(\mathbf{z}, C)$  to approximate the expected outcome utility. To remove the confounding bias where frequent paths are automatically assumed correct, we train  $V_\phi$  using an Inverse Propensity Score (IPS) objective over an offline dataset of trajectories  $\mathcal{D}_E$ :

$$\mathcal{L}_{\text{IPS}}(\phi) = \mathbb{E}_{(\mathbf{z}, Y) \sim \mathcal{D}_E} \left[ \frac{1}{\max(\Pi_{\text{ref}}(\mathbf{z}|C), \xi)} (V_\phi(\mathbf{z}, C) - Y)^2 \right] \quad (5)$$

where  $\xi > 0$  is a clipping parameter to prevent extreme variance under highly rare trajectories.

**Epistemic Value-Gating (EVG).** To avoid contrarian degradation when the majority path is optimal, we introduce an adaptive gating function  $g(C)$  that measures our certainty regarding the sub-optimality of the majority track. Let  $\mathbf{z}_{\text{maj}} = \arg \max_{\mathbf{z}} \Pi_{\text{ref}}(\mathbf{z}|C)$  be the dominant strategy sequence. We define the Epistemic Value Gate as:

$$g(C) = \sigma \left( \frac{\max_{\mathbf{z}} V_{\phi}(\mathbf{z}, C) - V_{\phi}(\mathbf{z}_{\text{maj}}, C)}{\gamma \cdot \text{Var}_{\phi}[V_{\phi}(\mathbf{z}_{\text{maj}}, C)] + \epsilon} \right) \quad (6)$$

where  $\sigma$  is the sigmoid function,  $\text{Var}_{\phi}$  represents the epistemic uncertainty of our value network, and  $\gamma$  is a scaling hyperparameter. If the majority path is highly effective, or if our value model has high uncertainty regarding its failure,  $g(C) \rightarrow 0$ .

Our final test-time trajectory reweighting operator  $W(\mathbf{z})$  modifies the selection probability of a reasoning trajectory during decoding or search planning:

$$W(\mathbf{z}) = \exp \left( \frac{\log \Pi_{\text{ref}}(\mathbf{z}|C) + g(C) \cdot \alpha \cdot V_{\phi}(\mathbf{z}, C)}{\tau_{\text{temp}}} \right) \quad (7)$$

where  $\alpha$  is the utility amplification factor and  $\tau_{\text{temp}}$  is the decoding temperature.

#### 4.4 Algorithm and Pseudocode

Algorithm 1 formally outlines the test-time search and trajectory reweighting loop of the CORE framework.

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##### Algorithm 1 CORE: Counterfactual Outcome-Regularized Estimation & Search

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**Require:** Task Context  $C$ , Reference Policy  $\pi_{\text{ref}}$ , Counterfactual Value Network  $V_{\phi}$ , Latent Mapper

$\Psi$ , Sample Size  $N$ , Exploration Parameter  $\alpha$ , Temperature  $\tau_{\text{temp}}$

1: Sample  $N$  raw candidate trajectories from the reference policy:  $\{\tau^{(i)}\}_{i=1}^N \sim \pi_{\text{ref}}(\cdot|C)$

2: **for** each trajectory  $\tau^{(i)}$  **do**

3: Map raw tokens to Latent Semantic Decision Nodes:  $\mathbf{z}^{(i)} = \Psi(\tau^{(i)})$

4: Compute structural prior probability:  $\Pi_{\text{ref}}(\mathbf{z}^{(i)}|C) = \prod_k P_{\text{ref}}(z_k^{(i)}|z_{<k}^{(i)}, C)$

5: **end for**

6: Identify the empirical majority trajectory sequence:  $\mathbf{z}_{\text{maj}} = \arg \max_{\mathbf{z}^{(i)}} \Pi_{\text{ref}}(\mathbf{z}^{(i)}|C)$

7: Estimate expected utility values and epistemic variance using  $V_{\phi}$ :

8:  $\mu_{\text{maj}} = V_{\phi}(\mathbf{z}_{\text{maj}}, C)$ ,  $\sigma_{\text{maj}}^2 = \text{Var}_{\phi}(V_{\phi}(\mathbf{z}_{\text{maj}}, C))$

9: Compute Epistemic Value Gate:  $g(C) = \sigma \left( \frac{\max_i V_{\phi}(\mathbf{z}^{(i)}, C) - \mu_{\text{maj}}}{\gamma \cdot \sigma_{\text{maj}}^2 + \epsilon} \right)$

10: **for** each candidate trajectory  $i = 1 \dots N$  **do**

11: Compute unnormalized reweighted score:

12:  $\log S_i = \log \Pi_{\text{ref}}(\mathbf{z}^{(i)}|C) + g(C) \cdot \alpha \cdot V_{\phi}(\mathbf{z}^{(i)}, C)$

13: **end for**

14: Normalize scores via softmax to obtain selection distribution:  $P_{\text{CORE}}(i) = \frac{\exp(\log S_i / \tau_{\text{temp}})}{\sum_j \exp(\log S_j / \tau_{\text{temp}})}$

15: **return**  $\tau^{(k)}$  where  $k = \arg \max_i P_{\text{CORE}}(i)$

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## 5 Theoretical Analysis & Proof of Safety

We provide a theoretical guarantee that our objective prevents performance degradation under standard regimes where the majority prior is already aligned with the optimal outcome.

**Theorem 1** (Bounded Regret and Consistency Safety). *Let  $\tau^* = \arg \max_{\tau} U(\tau)$  be the optimal trajectory. Assume that the majority reference trajectory is optimal, meaning  $\mathbf{z}_{\text{maj}} = \Psi(\tau^*)$ . Let  $\mathcal{R}(\pi) = U(\tau^*) - \mathbb{E}_{\tau \sim \pi}[U(\tau)]$  define the expected regret of a policy  $\pi$ . Under perfect value calibration up to epistemic bound  $\epsilon$ , the regret of the **CORE** selection policy  $\pi_{\text{CORE}}$  is strictly bounded by the reference policy regret up to an arbitrarily small slack variable:*

$$\mathcal{R}(\pi_{\text{CORE}}) \leq \mathcal{R}(\pi_{\text{ref}}) + \mathcal{O}(\epsilon) \quad (8)$$

*Proof.* By definition, if  $\mathbf{z}_{\text{maj}} = \Psi(\tau^*)$ , then  $\max_{\mathbf{z}} V_{\phi}(\mathbf{z}, C) = V_{\phi}(\mathbf{z}_{\text{maj}}, C) \pm \epsilon$ . Substituting this into the Epistemic Value Gate numerator:

$$\Delta V = \max_{\mathbf{z}} V_{\phi}(\mathbf{z}, C) - V_{\phi}(\mathbf{z}_{\text{maj}}, C) \leq \epsilon \quad (9)$$

As  $\epsilon \rightarrow 0$ , the numerator approaches zero or a negative residual bounded by value resolution. Thus,  $g(C) = \sigma(\frac{\Delta V}{\text{variance}}) \rightarrow \sigma(0) = 0.5$  if exactly identical, or rapidly collapses to 0 if  $\mathbf{z}_{\text{maj}}$  is strictly greater than all minority alternatives by more than the noise threshold. When  $g(C) \rightarrow 0$ , the reweighting exponent simplifies directly to:

$$W(\mathbf{z}) \propto \Pi_{\text{ref}}(\mathbf{z}|C)^{1/\tau_{\text{temp}}} \quad (10)$$

which preserves the precise rank-ordering of the reference policy distribution  $\pi_{\text{ref}}$ . Hence, the probability of choosing a sub-optimal contrarian trajectory approaches zero, bounding the operational regret to that of the base distribution matching paradigm.  $\square$

## 6 Experimental Setup

We execute a comprehensive empirical validation of **CORE** against leading edge models and search algorithms across two primary axes: Advanced Hard Reasoning (Mathematics/Science) and Long-Horizon Agent Execution.

### 6.1 Benchmarks

- **MATH & AIME**: Standard and Olympic-level multi-step mathematical reasoning datasets demanding exact symbolic correctness [21].
- **GPQA**: Graduate-level Google-Proof Q&A benchmark covering physics, chemistry, and biology, highly vulnerable to plausible-sounding majority errors [36].
- **WebArena**: A highly complex, long-horizon web-agent environment where actions involve structural HTML interaction, web-shopping, and database configuration over hundreds of steps [48].
- **SWE-bench**: Real-world software engineering agent tasks requiring long-horizon code analysis, file modification, and passing specific test suites [24].
- **GAIA**: General AI Assistants benchmark tracking multi-modal, long-horizon tasks requiring tool orchestrations and precise pipeline execution [30].

## 6.2 Baselines

We compare **CORE** against a comprehensive array of contemporary paradigms: 1. **Chain-of-Thought (CoT) Greedy**: Direct decoding from a standard base reasoning policy [45]. 2. **Self-Consistency (SC@64)**: Sampling 64 paths and extracting the majority vote answer [44]. 3. **Outcome Reward Model (ORM)**: Trajectory selection guided by a standard final-state verifier. 4. **Process Reward Model (PRM)**: Step-by-step verification search using a state-of-the-art token verifier trained without IPS correction [29]. 5. **DeepSeek-R1-Style Tree Search**: An advanced test-time search mechanism maximizing standard expected reward without majority bias correction [14]. 6. **Implicit Q-Learning (IQL)**: Standard offline RL sequence baseline that maximizes reward while tightly regularizing back to  $\pi_{\text{ref}}$  [26].

## 7 Main Results and Discussion

Table 1 summarizes the performance of **CORE** against baseline paradigms across all primary evaluation metrics.

Table 1: Main Results across Mathematical Reasoning and Long-Horizon Agent Benchmarks. Figures report exact-match accuracy (%) for MATH, AIME, and GPQA, and task success rate (%) for WebArena, SWE-bench, and GAIA. Best results are bolded.

| Method                    | MATH        | AIME        | GPQA        | WebArena    | SWE-bench   | GAIA (L3)   |
|---------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| CoT Greedy                | 68.4        | 18.2        | 42.1        | 14.3        | 11.2        | 28.4        |
| Self-Consistency (SC@64)  | 74.2        | 22.4        | 45.6        | 16.1        | 12.8        | 31.2        |
| ORM Selector              | 76.8        | 25.1        | 48.3        | 19.4        | 15.3        | 34.9        |
| PRM Tree Search           | 82.3        | 34.5        | 54.2        | 22.8        | 19.1        | 41.5        |
| DeepSeek-R1-Style MCTS    | 85.1        | 39.2        | 58.7        | 26.5        | 22.4        | 45.1        |
| Offline RL (IQL) Baseline | 78.5        | 28.3        | 50.1        | 20.2        | 16.8        | 36.4        |
| <b>CORE (Ours)</b>        | <b>91.4</b> | <b>52.6</b> | <b>71.3</b> | <b>38.9</b> | <b>31.7</b> | <b>58.2</b> |

### 7.1 Analysis of Performance Gains

As demonstrated in Table 1, **CORE** significantly outperforms all distribution-matching and traditional test-time scaling baselines. On exceptionally difficult mathematical domains like **AIME**, where the correct logical path is frequently counter-intuitive and constitutes an extreme minority branch, **CORE** improves accuracy from the MCTS baseline of 39.2% to **52.6%**. This absolute gain of 13.4% highlights the critical limitation of traditional search engines that rely on standard PRMs, which routinely filter out rare but brilliant mathematical insights due to low generation priors.

In long-horizon agent environments like **WebArena** and **SWE-bench**, where multi-step error propagation causes standard models to loop into repetitive high-frequency fallback sequences, **CORE** successfully activates the Epistemic Value Gate, recognizes the majority failure mode, and switches to clean, lower-frequency alternative execution pathways.

## 8 Ablation Studies and Diagnostic Analyses

We conduct rigorous control experiments to dissect the functional source of **CORE**’s performance gains and validate our safety assertions.



## 8.1 Ablation of Key Framework Elements

We systematically remove components of the **CORE** architecture to observe their impact on performance. Table 2 details the outcomes on **MATH** and **WebArena**.

Table 2: Ablation study evaluating structural dependencies in the **CORE** framework.

| Configuration                                         | MATH (Accuracy %) | WebArena (Success %) |
|-------------------------------------------------------|-------------------|----------------------|
| <b>Full CORE Framework</b>                            | <b>91.4</b>       | <b>38.9</b>          |
| <i>w/o</i> Latent Semantic Mapping (Token-level)      | 81.2              | 24.6                 |
| <i>w/o</i> Inverse Propensity Weighting (IPS)         | 83.5              | 27.2                 |
| <i>w/o</i> Epistemic Value-Gating ( $g(C) = 1$ fixed) | 74.8              | 21.3                 |

The ablation reveals that removing the Epistemic Value-Gating mechanism causes severe performance drops (**74.8%** on **MATH**). This directly confirms that without the adaptive gate, a model modified to pursue minority tracks will blindly choose erratic, lower-frequency paths on simple problems where the majority prior was completely accurate.

## 9 Case Studies and Qualitative Error Analysis

### 9.1 Mathematical Reasoning Example (AIME Problem)

**Context:** Find the number of ordered pairs of integers satisfying a non-linear system of modular equations.

**Majority Trajectory ( $\mathcal{T}_{\text{HF-LO}}$ ):** The agent attempts standard algebraic substitution. This approach is highly frequent in the training set for algebraic equations. However, it leads to an infinite expansion of complex terms that cannot be factored within token limits, resulting in a guessed final answer of 0 or 1 (Observed prior probability: 0.84, True Utility: 0).

**CORE Selected Trajectory ( $\mathcal{T}_{\text{LF-HO}}$ ):** The model maps the state to a latent node representing a rare application of the Chinese Remainder Theorem combined with a geometric lattice boundary condition. This trajectory has a very low reference prior (0.02) but is assigned a massive causal utility score by  $V_\phi$ . **CORE** scales its weight up, allowing the model to complete the elegant proof successfully.

### 9.2 Failure Modes & Limitations

Despite its strengths, **CORE** faces specific limitations:

1. **Extreme Sparsity Bounds:** If the optimal path is so rare that it is never generated within the  $N$  sample rollouts, **CORE** cannot score it. It selects the best available path in the sample space, falling back to a sub-optimal alternative.
2. **Out-of-Distribution Value Instability:** If the latent mapping  $\Psi$  encounters an entirely novel semantic action sequence, the value model  $V_\phi$  can produce erratic utility estimates, though this is heavily buffered by the epistemic variance term in the gate.



## 10 Conclusion and Future Work

In this paper, we introduced **CORE**, a framework that transitions language agents from historical distribution matching to proactive future outcome optimization. By formalizing the structural differences between high-frequency, low-outcome paths and low-frequency, high-outcome paths, we address a fundamental blind spot in standard generative AI alignment. Through Latent Semantic Abstractions and Epistemic Value-Gating, **CORE** eliminates majority-frequency bias on complex reasoning and long-horizon tasks without degrading performance when common-sense defaults are correct. Future work will focus on integrating the counterfactual estimation objective directly into online, recursive test-time reinforcement learning loops to synthesize high-utility minority paths dynamically.

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## A Appendix: Hyperparameters and Extended Implementation Details

### A.1 Implementation and Computational Budget

Our experiments utilize a base policy fine-tuned from Llama-3-70B-Instruct. The Counterfactual Value Network  $V_\phi$  is configured as an auxiliary multi-layer perceptron (MLP) branch mapped onto the pooled transformer representations at layer 32, containing 3 training blocks of width 2048 with SwiGLU activations. For the Latent Semantic Decision Node mapping  $\Psi$ , we utilize a discrete codebook quantized via VQ-VAE fine-tuned specifically on agent structural interaction syntax.

Table 3 presents the hyperparameter configurations utilized to ensure reproducible execution across all evaluated benchmarks.

Table 3: Hyperparameter Specifications for CORE Execution.

| Hyperparameter           | Description                                   | Value              |
|--------------------------|-----------------------------------------------|--------------------|
| $N$                      | Number of trajectory generation samples       | 64                 |
| $\alpha$                 | Causal utility amplification factor           | 2.4                |
| $\gamma$                 | Epistemic variance gate scaling weight        | 1.5                |
| $\tau_{\text{temp}}$     | Generation softmax decoding temperature       | 0.7                |
| $\xi$                    | Inverse Propensity Score lower clipping bound | 0.05               |
| $\epsilon$               | Numerical stability constant                  | $1 \times 10^{-5}$ |
| Optimizer                | Weight update optimization method             | AdamW              |
| Learning Rate ( $\phi$ ) | Value model learning velocity                 | $3 \times 10^{-5}$ |
| Batch Size               | Context batching dimensionality               | 128                |

### A.2 Extended Validation: Real-World Verification Shift

To examine if CORE suffers under dynamic real-world environment drift, we subjected our agent framework to an evaluation over a scrambled version of WebArena where background server execution paths were shifted intentionally to yield high-frequency human dead ends. Under this critical distribution adversarial drift, standard PRMs collapsed completely in action success rates to below 5%, while CORE maintained a resilient success rate of 34.2% due to the strict IPS un-biasing properties of the counterfactual framework.